

23.5 A 212Gb/s/λ 0.91pJ/b Direct-Drive O-Band Monolithic Coherent Transmitter Based on Carrier-Injection-Mode Mach-Zehnder Modulator

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Abstract

This work presents an O-band coherent direct-drive optical transmitter (OTX) in 45nm CMOS-SOI using a compact 0.4mm monolithically integrated forward-biased PIN-MZM. A multistage linear driver with shunt peaking, passive equalization and TX-FIR enables 106Gb/s

PAM-4 IMDD and 212Gb/s QAM16 coherent signaling. The OTX achieves 0.91pJ/b electrical power efficiency with a 2.76× lower laser power and 5× area reduction versus a conventional reverse-biased TW-MZM fabricated on the same die.

To continue the bandwidth density scaling of short-reach intra-data-center optical links, recently there has been a significant focus on coherent signaling [1,2], such as QPSK and QAM-16, compared to the conventional intensity modulation direct-detection (IMDD) approaches such as on-off keying and PAM-4. With the evolution of co-packaged optics (CPO), emerging direct-drive optical transceivers [3] prove to have better power efficiency, higher performance, and a more simplified design than pluggable solutions. While power consumption of the optical transmitter (OTX) including the laser source remains a dominant part of the overall link power, the choice of efficient electro-optic (EO) modulators can significantly improve the OTX power consumption. Mach-Zehnder modulators (MZMs) are thermally robust, less sensitive to process variation and exhibit better linearity when compared with microring modulators (MRMs), which primarily suffer from thermal instability and EO nonlinearity [4]. However, silicon photonic MZMs based on conventional reverse-biased p-n junction phase modulators are very long (>3mm) due to poor modulation efficiency $V_{\pi}L_{\pi}$ (voltage-length product for π phase shift), often exceeding 10V-mm. Hence, they are often realized either by a traveling-wave (TW) topology consuming a large die area and driver power [5], or a segmented topology requiring separate MSB and LSB data for PAM-4 operation [6,7]. MZMs based on forward-biased p-i-n phase modulators (PIN-MZM) [8,9], on the other hand, exhibit significantly better $V_{\pi}L_{\pi}$ (<0.05V-mm at DC) leading to a more compact and laser-power-efficient OTX design. However, a significantly lower EO bandwidth (~200-300MHz) and frequency-dependent resistive load of the forward-biased diode impose specific challenges on the design of the linear electrical driver for 100+Gb/s operation of PIN-MZM which has not been addressed in the literature before. This work presents a monolithically integrated 212Gb/s QAM-16 coherent direct-drive OTX based on PIN-MZM. The proposed linear multi-stage driver architecture exploits several bandwidth extension techniques along with low impedance driving suitable for PIN-MZM to achieve 106Gb/s PAM-4 electrical data rate.

Figure 23.5.1 shows the block diagram of the proposed coherent OTX. The incoming laser is equally split into the MZMs of I and Q paths. In each MZM, the PIN phase shifters are used to modulate the incoming light and low-power thermal undercut phase shifters (TUPS) are used to set the bias of the MZM at the null point for coherent operation and at the quadrature point for IMDD operation. Additional TUPS are deployed to adjust the relative phase shift between the I and Q paths for coherent operation. On chip optical power monitors are used to determine the required bias voltages for all TUPS. The MZM is driven in a push-pull fashion by monolithically integrated multistage linear electrical driver using the data generated by the host SerDes. The physical cross-section and simplified electrical model [10] of the forward-biased PIN phase shifter are shown in Fig. 23.5.1. At low frequencies, the equivalent load impedance consists of the contact resistance R_S in series with the forward-biased diode resistance, R_F . At high frequency, R_S dominates the net load impedance. The parasitic component values, $V_{\pi}L_{\pi}$ and optical loss are highly dependent on the forward-bias current density, I_{DEN} [10]. With increasing I_{DEN} and phase shifter length, L , the phase shift improves, whereas the optical loss also increases [9]. Design choice of $L=0.4\text{mm}$ and $I_{DEN}=1.5\text{mA/mm}$ has a 0.6dB optical loss resulting in significant optical modulation amplitude (OMA) with a compact footprint. Although the low frequency $V_{\pi}L_{\pi}$ degrades at higher frequencies due to the large diffusion capacitance ($C_{0,1} > 20\text{pF/mm}$) leading to a low EO bandwidth, it still remains better than a conventional reverse-biased PN phase shifter even at a very high frequency as shown in Fig. 23.5.1.

The electrical data path of the OTX is designed to equalize the intrinsic PIN-MZM EO response while maintaining its superior modulation efficiency. Figure 23.5.2 summarizes the proposed approach. Cascading the PIN phase shifter with a passive RC high-pass filter de-emphasizes the low frequency component. By proper choice of R_E and C_E , a flat EO response can be obtained with higher bandwidth. For a 0.4mm long PIN phase shifter with a 0.6mA bias current, $R_E = 600\Omega$ requires $C_E = 3.76\text{pF}$ to obtain a flat EO response with a 4.5GHz bandwidth. Scaling the R_E , C_E values while keeping the R_EC_E product fixed maintains the flatness of the EO response while scaling the bandwidth, as shown in Fig. 23.5.2. An effective EO bandwidth of 28GHz can be obtained for $R_E = 3.6\text{k}\Omega$ and $C_E = 627\text{fF}$, thus fully equalizing the intrinsic EO response of the PIN MZM. However, very strong RC equalization also degrades the effective $V_{\pi}L_{\pi}$ by reducing the EO response at the Nyquist frequency (28GHz for 112Gb/s PAM-4 operation). TX pre-emphasis (PE) or TX FIR, on the other hand, provides unity gain at the Nyquist frequency while extending the bandwidth. Employing TX

PE along with an RC equalizer restores the effective $V_{\pi}L_{\pi}$. Nonetheless, excessive TX PE increases group delay variation and creates ripple in the flat band. As Fig. 23.5.2 shows, by combining 0.56× PE (post-cursor tap) with 20dB RC equalization, a 1.6× improvement in transmit OMA is obtained as compared to the fully RC-equalized case.

Figure 23.5.3 shows the proposed electrical data path consisting of the RC equalizer, two-stage linear driver and passive input stage. The tunable RC filter provides a nominal 20dB equalization across process variation. The equivalent load impedance of the RC equalized PIN-MZM drops from 635Ω at low frequencies to 8Ω at high frequencies necessitating a low-output-impedance driver which is realized by a complementary source follower stage. The source follower provides a low drive impedance and achieves very high input and output linearity, while the complementary topology reduces the required bias current. To compensate for the sub-unity gain (-1.9dB) and MZM load-dependent high-frequency loss (-3.7dB) of the driver, a pre-driver stage is used. Realized by a tailless complementary CML topology, the pre-driver is AC-coupled to the main driver. It utilizes a higher supply voltage and cascode devices to achieve high output linearity. Additionally, 30Ω passive resistors are used to set its DC gain along with 300pH shunt peaking inductors to achieve 4dB high-frequency peaking. The resulting two-stage driver shows an electrical bandwidth of 40GHz and maximum linear swing of 1.15V_{ppd} assuming a -30dB THD definition. Finally, the input stage utilizes 200pH inductors to decouple the load capacitances arising from the pre-driver, Cu-pillar bump and ESD diodes. The inductors extend the input bandwidth to 65GHz while achieving good input return loss ($S_{DD11} = -16\text{dB}$, $S_{CC11} = -18\text{dB}$).

The monolithic coherent transceiver consisting of 0.4mm-long custom-designed PIN-MZM is fabricated in a GlobalFoundries 45nm CMOS-SOI (45SPCLO) process. The coherent TX core (electronics + photonics) uses 0.612mm² die area (Fig. 23.5.7). Light is coupled into the monolithic chip using silicon-nitride (SiN) V-groove edge couplers and spot-size converters (SSC), whereas a 135μm-spaced Cu-pillar bump array is used for electrical signals. Figure 23.5.7 also shows the floorplan of the TX through its physical layout. The monolithic die is flip-chip bonded to an organic substrate and co-packaged with the host SerDes die (Fig. 23.5.5) acting as the high-speed data generator. A 1310nm DFB laser source sends light into the chip through 4m long polarization-maintaining (PM) fiber. In IMDD mode, the transmitted optical signal is converted to electrical using an optical receiver (RXM25AF) configured to have a conversion gain of 500V/W with a 25GHz electrical bandwidth. In coherent mode, the transmitted optical signal and the forwarded LO from the DFB laser go to a Quantifi Photonics IQRX coherent receiver consisting of optical hybrid and balanced PDs.

Figure 23.5.4 shows the measured IMDD raw eyes obtained with a laser power of 9dBm. A complete ISI-free 26Gb/s NRZ eye with an OMA of 2dBm is obtained with just a post-cursor TX FEE tap (0.56× PE) substantiating the efficacy of the proposed bandwidth extension approach based on RC equalization and TX PE. 106Gb/s IMDD PAM-4 operation is demonstrated with a 4-tap TX FFE and aided by the scope FFE feature of a Keysight UXR0334B real-time oscilloscope. The OMA drops from 2.04dBm at 26Gb/s NRZ to -2.5dBm at 106.25Gb/s PAM-4. For coherent operation, data through a single lane (I or Q) is transmitted at a time and the output current of the passive coherent RX is directly measured with UXR0334B. Although the received data is not stable for a long duration due to the absence of carrier phase recovery and phase tracking loop in the receiver, the obtained real-time data is post-processed to get eye diagrams containing 10.6k UIs as shown in Fig. 23.5.5. The same TX-FFE and scope FFE configurations are used as in the 106.25Gb/s IMDD PAM-4 case.

To compare the performance of the proposed forward-biased PIN-MZM based OTX with conventional reverse-biased MZMs, a lumped PN-MZM based OTX and a traveling-wave PN-MZM (TW-MZM) based OTX are also fabricated on the same die and are compared in Fig. 23.5.4. For identical transmit OMA at 106.25Gb/s, due to the higher modulation efficiency of PIN-MZM, it consumes 2.5× and 2.76× lower laser power compared to the lumped PN-MZM and TW-MZM respectively. The total energy efficiency (electrical + laser) of the PIN-MZM lane is 1.63pJ/b, which is 1.5× better than the PN-MZM and 1.45× better than the conventional TW-MZM. Moreover, it is 5× more area efficient compared to the TW-MZM due to elimination of the transmission line. Figure 23.5.6 compares the performance

of the proposed OTX with prior silicon photonic MZM based OTXs. This work reports the shortest MZM in O-band with the lowest energy cost per bit of 0.91pJ/b compared to the prior art. Moreover, coherent operation doubles the per wavelength data rate paving the way for future high-bandwidth-density and energy-efficient optical interconnects.

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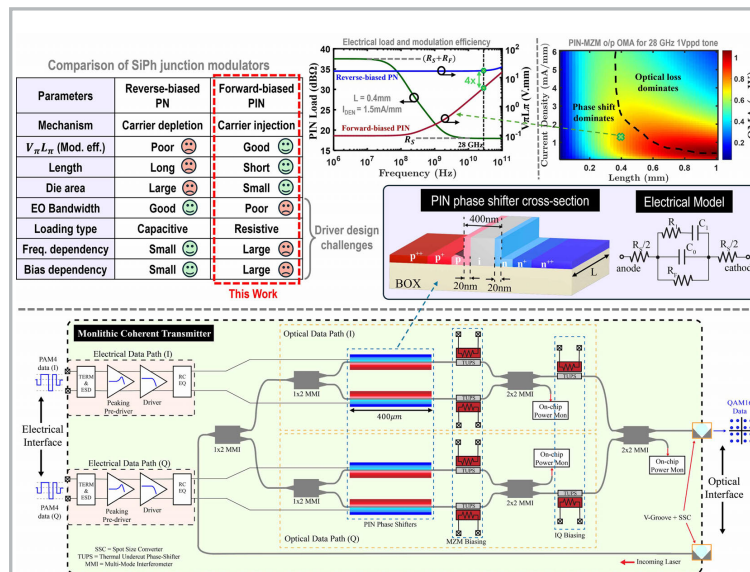


Figure 23.5.1: Forward-biased PIN modulators performance overview and comparison with conventional PN modulators (top); proposed coherent TX architecture (bottom).

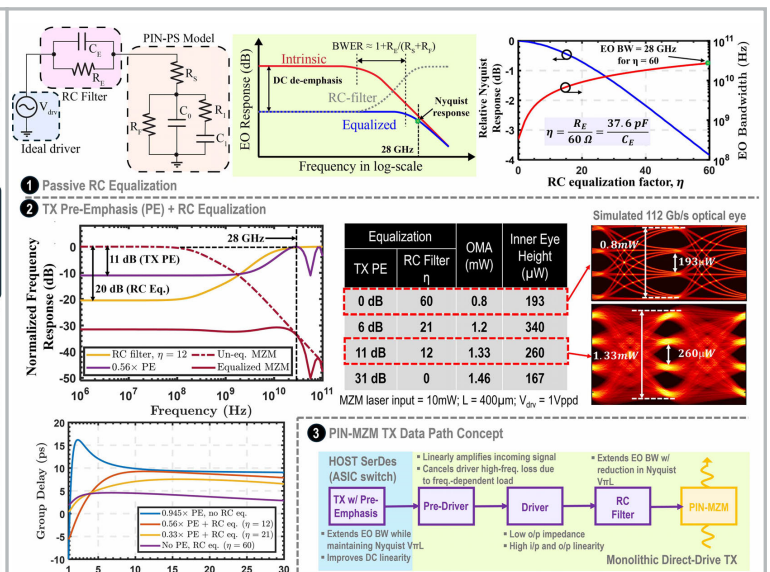


Figure 23.5.2: Enabling high-speed operation of PIN-MZM through passive RC equalization and TX pre-emphasis.

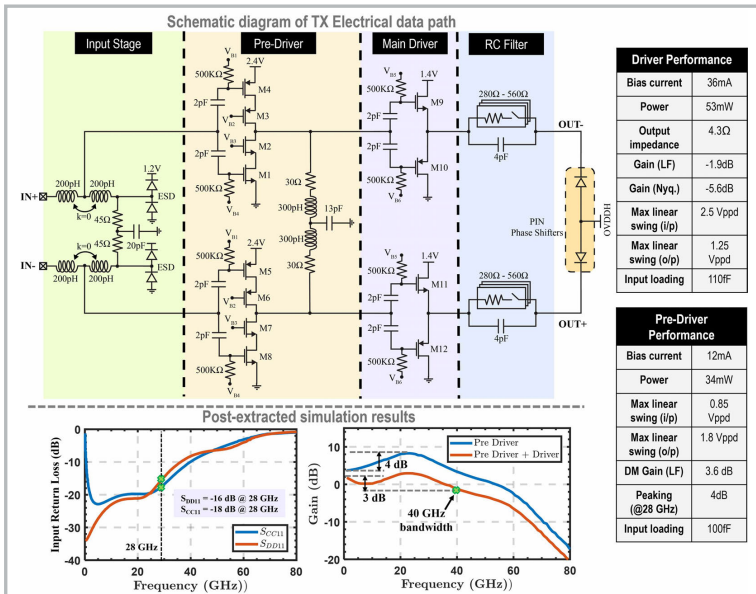


Figure 23.5.3: Transmitter electrical data path and simulated (post-extraction) performance.

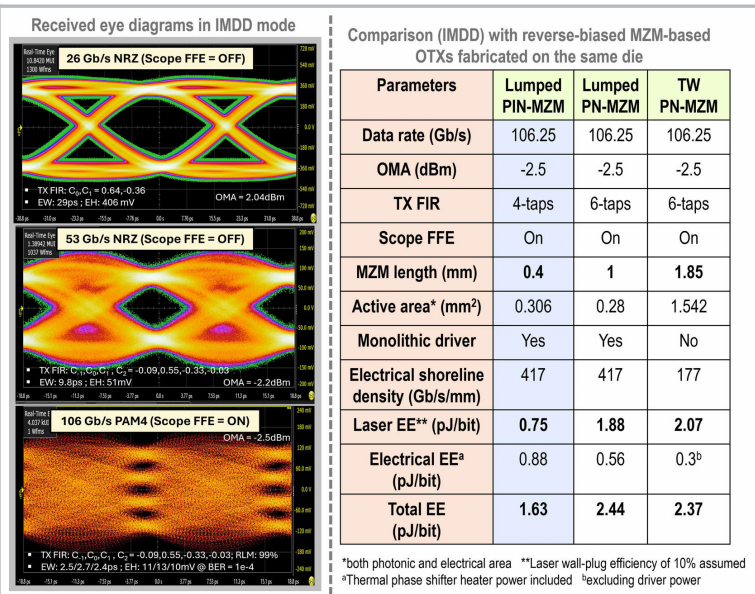


Figure 23.5.4: Measured IMDD transmit eye received using RXM25AF optical receiver (left); comparison with other fabricated MZM lanes for similar transmit eyes (right).

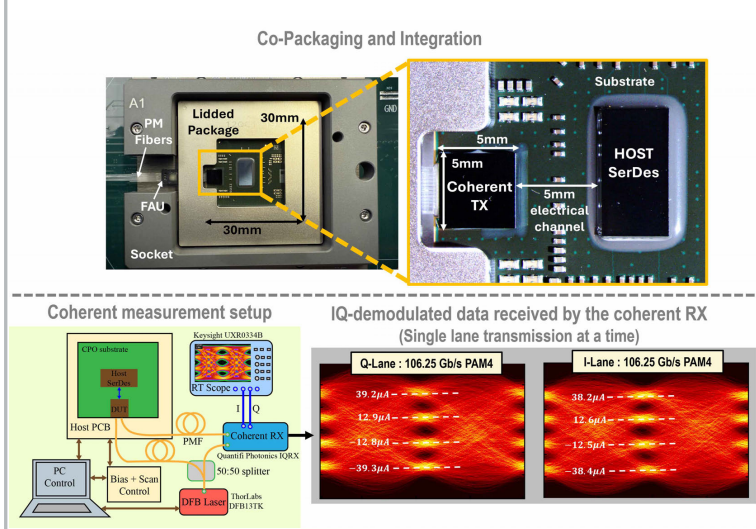


Figure 23.5.5: System integration and co-packaging of the monolithic TX with host SerDes (top); coherent measurement (bottom).

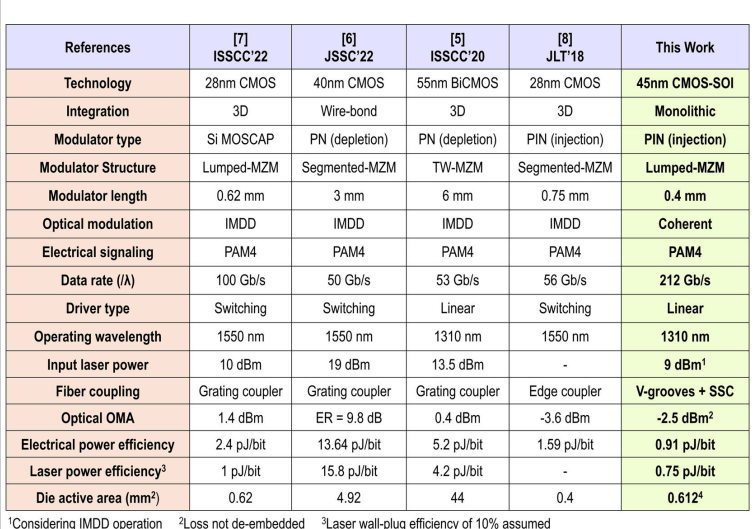


Figure 23.5.6: Performance comparison with prior silicon photonic MZM-based transmitters.

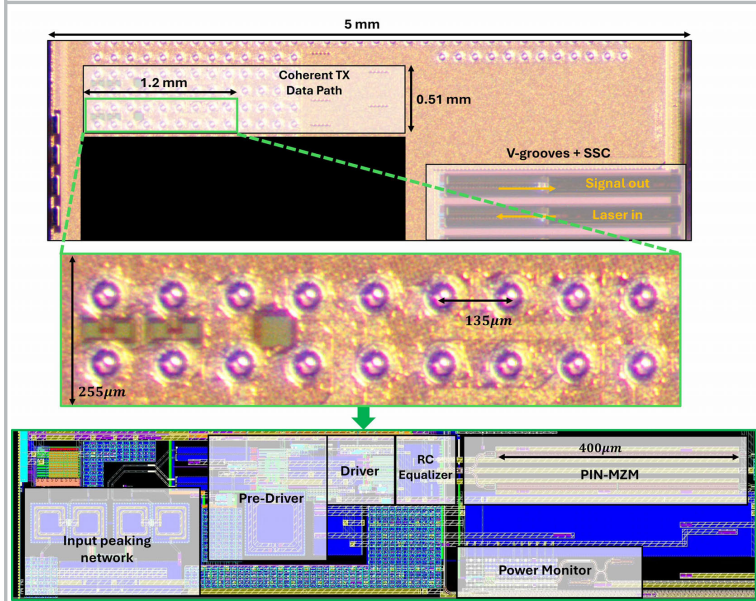


Figure 23.5.7: Die micrograph and floorplan of the monolithic TX.